

Abdulrahman Eldeeb

Python Developer

📍 Fayoum, Egypt | 📞 (+20) 102-671-9792 | ✉️ ar2724@fayoum.edu.eg
🌐 [linkedin.com/in/abd-el-rahman-eldeeb](https://www.linkedin.com/in/abd-el-rahman-eldeeb) | 🐙 github.com/ElDEEB21

Summary

Computer Science graduate and Codeforces Expert focused on backend development with Python – Django, Django REST Framework, and FastAPI. Experienced in building production-ready RESTful APIs using Dockerized workflows, JWT authentication with Redis-backed revocation, Celery background tasks, automated testing, and OpenAPI documentation. Strong foundation in algorithms and competitive programming.

Professional Experience

Instructor

Sep 2024 – Aug 2026

Recode Academy

Fayoum, Egypt

- Taught multiple courses including CS50, C++ programming, algorithms and data structures, and Object-Oriented Programming (OOP) core concepts.
- Delivered structured curriculum covering problem-solving fundamentals, computational thinking, and advanced programming paradigms to diverse student cohorts.

Mentor & Coach

April 2023 – Mar 2025

Fayoum ICPC Community

Fayoum, Egypt

- Mentored 15–20 students in programming fundamentals, data structures, and algorithm techniques to prepare for the Egyptian Collegiate Programming Contest (ECPC).
- Organized programming workshops, training sessions, and coding competitions to enhance programming abilities of community members.

Projects

Bookly - Book Management API | FastAPI, SQLAlchemy, PostgreSQL, Redis, Celery Aug 2026 – Sep 2026

- Built an async book catalog API with FastAPI + SQLAlchemy covering JWT auth with refresh rotation, **Redis blocklist** for revoked tokens, role-based access, and owner-scoped book CRUD with reviews and tags.
- Learned **Celery** and implemented **background tasks** for email delivery – verification, resend-verification, and password-reset flows via FastAPI-Mail with signed time-limited tokens.
- Hardened the API with Redis **rate limiting** on auth endpoints, Alembic migrations, centralized error codes, and auto-generated Swagger/ReDoc docs.

E-Commerce REST API | Django REST Framework, PostgreSQL, Stripe, JWT, Redis Feb 2026 – Jun 2026

- Built a production-ready e-commerce API covering user auth, products, cart, orders, and **Stripe** payment processing with webhook-driven status updates.
- Applied **Service Layer & Selector Layer** patterns separating business logic from views and data access across 5 modular Django apps.
- Implemented **JWT auth** with token rotation, blacklisting, HTTP-only cookies, and split settings (base/dev/prod) with auto-generated Swagger docs.
- Added **Redis caching** with tiered TTLs, service-layer cache invalidation on writes, and environment-specific configurations.

Recipe API | Django REST Framework, PostgreSQL, Docker, OpenAPI, TDD Mar 2026 – Apr 2026

- Built a production-style Recipe Management API with user registration, token authentication, profile endpoints, and strict ownership isolation for user data.
- Implemented full CRUD for recipes, tags, and ingredients with advanced filtering (by tags/ingredients), assigned-only queries, and multipart image upload endpoints.
- Applied test-driven development with comprehensive unit tests, health checks, DB readiness workflow, and Docker Compose setup for consistent local and deployment environments.

Open Source

- **Open-source contributor** – merged pull requests into community skill libraries for coding-agent delegation.
- Refactored seven duplicated relay event scanners into one parity-gated implementation, fixing an unbounded memory-growth defect and adding 21 recovery test cases ([PR #39](#)).
- Shipped implementer delegation skills for the Cline and GitHub Copilot CLIs – Node-built-ins-only relays with brief-on-stdin transport and plan/read-only safety, verified against the real CLI contracts and their CI gates ([PR #59](#), [PR #87](#)).

Education

Fayoum University	Fayoum, Egypt
<i>Bachelor of Computer Science and Artificial Intelligence</i>	<i>Oct 2022 – Jun 2026</i>
• Relevant Coursework: Object-Oriented Programming, Data Structures, Algorithms Design & Analysis, Operating Systems, Database Management Systems, Computer Architecture	

Technical Skills

Languages	Python, C++, SQL, HTML/CSS, JavaScript
Backend	Django, Django REST Framework, Django ORM, FastAPI, SQLAlchemy, Pydantic, Celery, Class-Based Views, ViewSets, Routers
Databases	PostgreSQL, SQLite, MySQL, Redis
APIs & Auth	RESTful APIs, Token Authentication, JWT (SimpleJWT, PyJWT), Redis Blocklist, Permissions, Throttling, Rate Limiting, Stripe Integration, Webhooks, Swagger/OpenAPI
Architecture	MVT, Service Layer, Selector Layer, Modular App Design, RBAC, Async Python, Background Tasks, Clean Architecture, Test-Driven Development (TDD)
Tools	Git, GitHub, GitHub Actions (CI/CD), Docker, Docker Compose, Alembic, Uvicorn, Flake8, Postman, pip, virtualenv, django-filter
Data & ML	PyTorch, TensorFlow/Keras, OpenCV, Scikit-Learn, Pandas, NumPy, Power BI
AI-Assisted Dev	Coding agents (opencode, Claude Code), Model Context Protocol (MCP), multi-agent systems, agent orchestration, skill/tool development for AI workflows
Other	Web Scraping (Beautiful Soup, Selenium), Jupyter Notebooks, Google Colab

Competitive Programming

Codeforces Expert – Demonstrated strong algorithmic problem-solving skills
Ranked 17th Place in 2023 ECPC Qualifications (out of 280 teams)
Ranked 90th Place in the 2023 ECPC Final (out of 280 teams)

Certifications

- **Build REST APIs with Django REST Framework and Python** – Shubham Sarda (Udemy), 13 hours. Focus: serializers, APIViews/Generic Views, ViewSets & Routers, permissions, auth (Basic/Token/JWT), throttling, filtering, pagination, and API testing with Postman.
- **Python Django - The Practical Guide** – Maximilian Schwarzmuller (Academind/Udemy), 23 hours. Focus: Django fundamentals, templates/static files, ORM relationships, forms, sessions, class-based views, file uploads, and deployment workflow.
- **Build a Backend REST API with Python & Django - Advanced** – Mark Winterbottom & Brooke Rutherford (Udemy), 15 hours. Focus: Docker, TDD, GitHub Actions CI/CD, Flake8 linting, custom user auth, image upload APIs, filtering/search, and production-oriented project structure.

Soft Skills

• Technical Training	• Team Leadership	• Coaching & Feedback
• Curriculum Development	• Problem-Solving	• Communication